

Specific Aims

We aim to establish the technical foundation for imaging-guided, patient-specific glaucoma surgeries. To achieve this, we will develop a robotic optical coherence tomography (OCT) to construct digital twins of the anterior segment of the eye. A personalized hydrodynamic model based on the digital twin will guide the selection of surgical locations in the aqueous humor outflow (AHO) pathway to optimize intraocular pressure (IOP) reduction.

Glaucoma is a leading cause of permanent blindness worldwide. The only proven treatment is IOP reduction. Elevated IOP arises because of increased resistance (or reduced facility) of the AHO pathways. Aqueous flows from the anterior chamber through the circumferential trabecular meshwork (TM), into Schlemm's canal (SC), through collector channels (CC) and aqueous veins to the episcleral veins. Medications can temporarily improve AHO facility. However, the pathological AHO resistance in glaucoma is progressive, and medications become less effective over time. Additionally, patient compliance remains a key limitation, ultimately leading to surgery.

Minimally invasive glaucoma surgeries (MIGSs) are safe and fast procedures to lower IOP; however, their efficacy yields few mmHg in IOP reduction in well-controlled clinical trials. Due to the lack of information on the physiology and pathophysiology of patient-specific AHO anatomy and the impact of trabecular procedures on AHO pathways, existing clinical decision-making for MIGSs is poorly defined. Most MIGS procedures are performed in the nasal quadrant due solely to easy access and without customization for individual patient factors that may improve IOP lowering. Because SC size, TM stiffness, and CC distributions are highly circumferentially heterogeneous and segmental, the ability to (a) obtain the detailed AHO pathway anatomy, (b) establish correlations between anatomical features and segmental functions, and (c) predict IOP reduction outcomes after interventions would potentially increase the efficacy of MIGS. A 3D digital twin of AHO pathway anatomy, linked to its segmental AHO function, would allow a hydrodynamic model to determine the optimal MIGS location for IOP reduction. However, no tool is currently available to noninvasively reconstruct such a digital twin.

We recently developed *in vivo* fluorescence aqueous angiography (AA), visualized circumferential AHO in patients, and found that AHO exhibited different segmental low- or high-AHO regions in glaucoma patients. We further showed that TM ablation at selected regions in the AHO pathway could double IOP reduction. However, AA is invasive, technically challenging, can cause unpredictable complications, and cannot provide 3D information. Anterior-segment OCT (AS-OCT) has the potential to satisfy our needs. However, existing AS-OCTs have limited field-of-view (FOV), axial resolution, and, more importantly, no flexibility to image the complex AHO pathway anatomy. The circumferential distance (>37 mm) and area (>200 mm²) of the AHO pathways are too large to fit in a single FOV. Meanwhile, the diameters of CCs and aqueous veins are typically 20-50 μ m, depending on the measurement direction, and the smallest SC height is less than 10 μ m, which cannot be well-resolved by existing AS-OCTs. An ideal AS-OCT would provide flexible circumferential imaging, maintain perpendicular illumination to the limbus to avoid projection artifact, and offer at least 5 μ m axial resolution.

We propose to develop a robotic AS-OCT with optimized spectral bandwidth to achieve high spatial resolution, flexible imaging, and, in principle, unlimited FOV to reconstruct the 3D digital twin and to tailor hydrodynamic analysis based on the digital twin to predict MIGS outcome. Our **long-term goal** is to improve clinical glaucoma treatment outcomes. Our **central hypothesis** is that 3D digital twin-based, patient-specific hydrodynamic analysis is the best way to achieve the most effective MIGS in each patient. The **objectives** are to build the necessary technical foundation with the following specific aims.

Aim 1: To develop, validate, and optimize a robotic AS-OCT platform. We will identify the optimal spectral range for satisfactory penetration depth and spatial resolution. We will develop a robotic AS-OCT for humans with the correct form factor to accommodate facial contour and safety features. We will create a safe operation protocol for the robotic AS-OCT and validate the protocol using 3D-printed dummy human heads with race, gender, and age variations following National Institute for Occupational Safety and Health (NIOSH) standards.

Aim 2: To reconstruct digital twins of the anterior segments and establish the anatomy/function correlation of segmental AHO. We will image 3D-printed eye phantoms, wild-type mice, SC fluorescent reporter (*Prox1-EGFP*) mice, and human donor eyes to reconstruct the digital twins of the anterior segments based on the simultaneous localization and mapping (SLAM) principle. We will map AHO pathway anatomical features (TM, SC, CCs, and aqueous veins) and correlate them with fluid flow information from AA imaging or blood- or lipid-emulsion-based AS-OCT angiography (AS-OCTA). We will further test the feasibility of continuously updating the digital twin after AHO perturbation, either pharmacologically or surgically.

Aim 3: To develop an eye-specific hydrodynamic model to guide MIGS. Based on the digital twin, we will develop an eye-specific hydrodynamic model for AHO. To test the model, we will perform AHO pathway laser ablation in mice and TM bypass in human donor eyes to induce segmental AHO alterations. We will validate and optimize the model-predicted IOP alterations with our intervention results and AA imaging.

At the end of the project, we will have developed and optimized the first robotic AS-OCT and establish a safe operation protocol to obtain the AHO digital twin. We will have developed and validated an eye-specific model based on the digital twin to predict IOP reduction. Simply put, we will have found the best ocular locations for MIGS that can optimally lower IOP.

RESEARCH STRATEGY

SIGNIFICANCE

There is an unmet need for clinical glaucoma treatment

Glaucoma care requires treatments that are safe, efficacious, and predictable. Glaucomatous optic neuropathy is a leading cause of irreversible blindness worldwide [1], and so far, no direct optic nerve treatment can reverse glaucoma. Thus, glaucoma treatment is risk factor modification. The only modifiable risk factor is elevated intraocular pressure (IOP) [2]. IOP reduction has been proven to mitigate glaucoma progression [3-6]. Yet current medical and surgical methods often fail to provide long-term IOP reduction to prevent blindness. Researchers attribute this to a lack of a comprehensive understanding of pathologic IOP in glaucoma.

IOP is the result of balancing aqueous humor production and aqueous humor outflow (AHO) [7]. The trabecular (or conventional) AHO pathways are the primary source of AHO resistance in the eye [8]. Classically, trabecular AHO has been approximated as a two-dimensional (2D) or linear depiction for IOP management. Aqueous humor is formed at the ciliary processes and then flows into the anterior chamber (AC). From there, aqueous passes through the trabecular meshwork (TM), into Schlemm's canal (SC), through collector channels (CC), and then into an intrascleral venous plexus that leads to aqueous veins (AV) and episcleral veins (EV) before returning to the systemic venous circulation. In glaucoma, elevated IOP is due to increased AHO resistance (or decreased AHO facility), which is primarily caused by pathological alterations at the TM [8, 9].

To reduce IOP, drugs (such as muscarinic agonists [10], rho-kinase inhibitors [11], or nitric-oxide donors [12]) and Minimally Invasive Glaucoma Surgeries (MIGSs) [13] target the TM to decrease AHO resistance. Unfortunately, pathological AHO resistance in glaucoma is progressive, and medications become less effective over time, leading to patients becoming medically non-responsive [14-17]. In addition, patient compliance remains a key limitation for medications, eventually leading to surgical interventions. MIGS are a quick and safe alternative to lower IOP by surgically ablating or bypassing pathological TM [13]. However, in large-scale, well-controlled clinical trials, average MIGS efficacy was lacking, providing only few mmHg of total IOP reduction [18, 19]. Therefore, improving MIGS efficacy remains a unmet clinical need.

MIGS efficacy suffers from the lack of knowledge of AHO anatomy and surgical planning

We argue that the lack of trabecular MIGS efficacy is caused by a poor understanding of the circumferential AHO anatomy and pathophysiology, which leads to a lack of surgical planning. Clinically, focal trabecular bypass and ablation in glaucoma patients near-universally target the TM in the nasal iridocorneal angle. The original justification was the concept of segmental trabecular AHO, where eyes have focal high-flow (HF) and low-flow (LF) regions around the limbus. Anatomical studies have shown a greater concentration of AHO pathways on the nasal side of the eye, and nasal MIGS were intended to access these HF pathways. However, no scientific evidence shows that HF-targeted surgery results in better clinical efficacy, individual AHO anatomy varies significantly among patients, and during early MIGS development, no clinical method exists to identify the segmental AHO status in any region in individual eyes. Therefore, the original concept of nasal targeting was not evidence-based and was not even guaranteed to access HF regions. In reality, the basis for nasal-targeted MIGS was the predominant use of temporal clear-cornea cataract surgery, where temporal cornea wounds used for cataract surgery become convenient for nasal MIGS and are more familiar to surgeons. In short, current MIGS do not involve surgical planning or address individual patient needs.

Personalized MIGS planning requires new, paradigm-shifting methodologies

To enable personalized MIGS planning, we need (1) the ability to image and quantify the fine anatomic details of the entire AHO pathway in each patient, (2) to re-create the AHO pathway anatomy as a virtual model, or "digital twin"; and (3) a robust AHO hydrodynamic model informed by the eye-specific digital twin to predict MIGS outcome. These three new abilities will collectively enable surgeons to determine the number and exact locations of trabecular bypass or ablation and predict the IOP reduction, all based on the digital twin and the hydrodynamic model. We believe such personalized operations will significantly improve MIGS outcome compared with the current nasal-targeting practice.

Among established imaging modalities, optical coherence tomography (OCT) is considered the gold standard for ocular imaging, providing high-resolution 3D visualization of fine anatomical features for managing nearly all blinding diseases [20, 21]. However, current anterior-segment OCT (AS-OCT) are not suited to image the AHO pathway for several reasons: (1) AHO pathways contain delicate structures, such as CCs that can be as small as 10 μm in diameter, which are deeply embedded ($\sim 50\text{-}500\text{ }\mu\text{m}$) within highly scattering scleral tissue. Imaging AHO requires a precise balance of center wavelength and bandwidth, which commercial AS-OCT systems fail to achieve. (2) The AHO pathways have a circular, post-limbal organization, with a circumferential distance exceeding 37 mm and an area over 200 mm^2 , both beyond the field of view (FOV) of current AS-OCTs. (3) These circular anatomical features prevent stationary imaging devices from maintaining normal (perpendicular) illumination along the entire AHO pathway. In OCT, normal illumination is crucial for minimizing optical path

length within the tissue, reducing light scattering, and maximizing OCT's inherent high axial resolution for accurate cross-sectional imaging. Hence, commercial AS-OCTs often require re-positioning the camera while patients adopt eccentric gazes to achieve proper alignment [30]. Therefore, existing AS-OCTs cannot feasibly image the circumferential AHO pathways; this is consistent with reported AHO AS-OCT studies where imaging is performed in small areas to attempt an extrapolated understanding of the entire AHO pathways [22-24].

To address the above technical challenges, we propose a robotic AS-OCT system with an unlimited FOV to visualize the circumferential AHO pathways with optimized spectral range to achieve an ideal balance between imaging depth and spatial resolution. During acquisition, AS-OCT illumination, controlled by a compact 6-degree-of-freedom (6-DoF) robotic arm, will remain normal to the limbus. We will reconstruct the digital twin of the AHO pathway from individually acquired AS-OCT volumes by developing a first-of-its-kind 3D montaging method. The digital twin will allow us to examine the AHO pathway in its physiological state and inform our recently developed hydrodynamic model [25] to be eye-specific. *Ultimately, the digital twin and tailored hydrodynamic model will allow surgeons to develop personalized MIGS plan to suit individual patient's needs for the first time.*

INNOVATION

The technology innovations include (1) developing a robotic AS-OCT with optimized spectral range, imaging depth, and spatial resolution for freeform human AHO pathway imaging; (2) creating the digital twin of AHO pathways for the first time using a new 3D montaging method; and (3) developing the first hydrodynamic model to predict MIGS outcome prospectively. The innovations in glaucoma investigation include (1) imaging the circumferential anatomy and glaucoma drug impact on the entire AHO pathways for the first time; (2) the first investigation of segmental AHO to create a structure/function relationship of AHO pathways in rodents and post-mortem human eyes; and (4) the first evaluation of personalized MIGS outcome.

RESEARCH TEAM

We assembled a multi-disciplinary team with complementary expertise to achieve our goals. The PI, Prof. Zhang, is a leader in OCT technology development and clinical translations for ophthalmic imaging. The MPI, Prof. Huang, is a clinician-scientist specializing in glaucoma care, MIGS, and AHO imaging. Prof. Sun specializes in robotics control and mechanical designs for OCT. Prof. Johnson is a leader in fluid dynamics in glaucoma and AHO. All investigators have established a productive collaboration for nearly ten years.

PRELIMINARY STUDIES

Imaging circumferential segmental AHO improves glaucoma treatment

We used aqueous angiography (AA) to image segmental AHO in human subjects (Figs. 1a-1e) [26] and treated glaucoma patients. For drug treatments, preliminary AA data shows that a muscarinic agonist (Miochol-E; acetylcholine; Ach 1%) improves focal low-flow areas to lower IOP in human subjects [27]. Baseline AA shows segmental AHO with low-flow (red arrows) and high-flow (green arrows) post-limbal regions (Fig. 1f). After application of Ach, only certain low-flow regions improve (Fig. 1g; orange arrows). For MIGS, we identify baseline low-flow regions (Fig. 1h; red arrows) in glaucoma patients and perform trabecular MIGS (Fig. 1h; blue arrows) [28]. Like drug treatment, MIGS targeting low-flow regions improves AHO (Fig. 1i; green arrows) [28]. Recently published perfusion studies add quantitative rigor and reproducibility, also showing that targeting trabecular bypass to baseline low-flow (as opposed to high-flow) regions of the AHO pathway quantitatively improves outflow facility to a greater extent (see Specific Aim 3) [29]. However, even if successful, AA itself is a time-consuming surgical procedure with associated infection, surgical, and anesthesia risks. In addition, as shown in Figs. 1e-1i, AA only provided an orthogonal projection view of the AHO pathway without any 3D information. Thus, a non-invasive 3D imaging modality, such as AS-OCT, would be better suited.

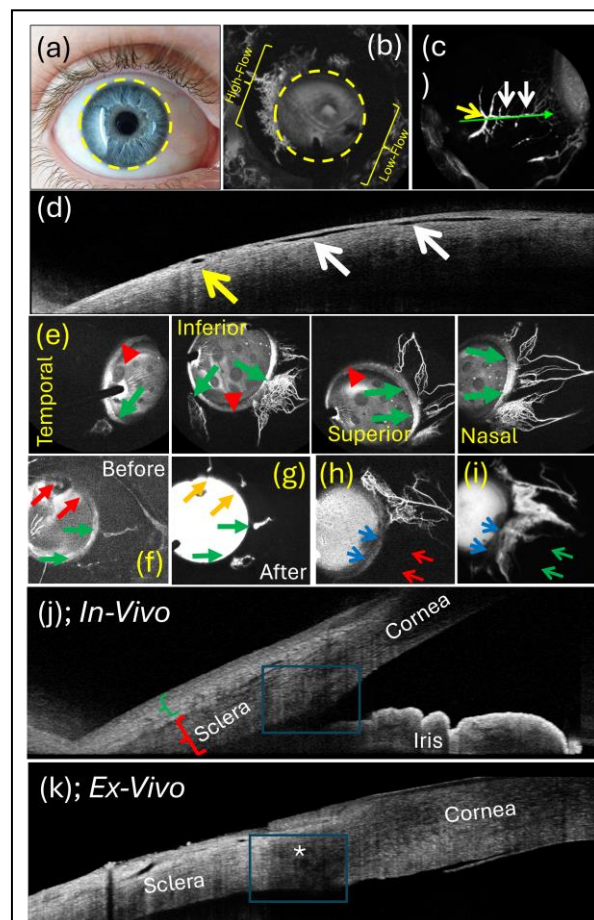


Fig. 1. Anterior segment AA and OCT. (a) Human eye. (b) Ex-vivo human eye showing segmental AHO from AA; (c) Human patient eye with AA signal (d) co-localizing with an OCT-imaged superficial aqueous vein. (Row e) Human patient eye showing segmental AHO. (f/g) Medical and (h/i) surgical rescue of low-flow regions. 840-nm AS-OCT in (j) a patient and (k) post-mortem human eye with shadowing (asterisk). Blue boxes: SC and AHO pathways are not clear.

Unfortunately, clinical AS-OCTs do not provide high-resolution comprehensive imaging of AHO pathways circumferential around the entire eye. Current AS-OCT illumination wavelength (840 nm) limits deeper imaging in patients. Superficial anatomy is well resolved (Fig. 1j; green bracket), but deeper regions (where many AHO pathways are located) are not (Fig. 1j; red bracket). In the lab, ex-vivo eyes are more difficult to study as significant shadowing arises from overlying conjunctiva or post-mortem tissue edema (Fig. 1k). Thus, a new AS-OCT must operate within a selected spectral range to achieve an optimal balance between imaging depth and axial resolution to image AHO pathways. Since commercial light sources (including superluminescent diodes and swept-source lasers) have limited or no tunable flexibility in the output spectra, we will develop robotic AS-OCT using a supercontinuum laser in this work to take advantage of its maximum spectral flexibility.

We developed supercontinuum OCT to circumferentially image AHO pathway anatomy in rodents

We developed a supercontinuum OCT, operating within the visible spectral range, to image circumferential AHO anatomy (SC and limbal microvasculature) noninvasively for the first time [30]. Our supercontinuum OCT, visible-light OCT or vis-OCT, operated between 510 nm and 610 nm, resulting in an axial resolution of 1.3 μm [30–32]. Vis-OCT provides much higher imaging contrast and sensitivity for imaging blood flow than near-infrared (NIR) OCTs due to increased optical scattering contrast and higher phase sensitivity for shorter wavelengths. High resolution and high contrast are critical for imaging rodent AHO anatomy [30]. Due to the smaller size of the mouse eyes, vis-OCT provides sufficient imaging depth.

Fig. 2 shows the 3D vis-OCT imaging of SC and venous plexus in wild-type mice. Fig. 2a is the 3D visualization of the cornea, iris, venous plexus, and SC. The venous plexus and SC are imaged by vis-OCT angiography (vis-OCTA). It is known that rodent SC can contain blood [33], which can then be increased with slight compression of the aqueous vein. As a result, these blood cells enhanced vis-OCTA contrast to image SC. Fig. 2b is a projection view of the vis-OCTA image. We can observe that SC is surrounded by a venous plexus and visualize their relationship with an aqueous vein. These *in-vivo* imaging results agree well with published labeled flat-mount results [34–36]. Figs. 2c & 2d are vis-OCT B-scan images along tangential and radial directions, which are respectively illustrated in Figs. 2e & 2f. We further acquired eight vis-OCT and vis-OCTA volume datasets around the eye by mechanically re-positioning vis-OCT scanning (Fig. 2g). Finally, we montaged the eight 2D projections and projected them onto a 3D sphere (Fig. 2h). These results build the foundation of our proposed robotic AS-OCT to image AHO anatomy as detailed in Aim 1.

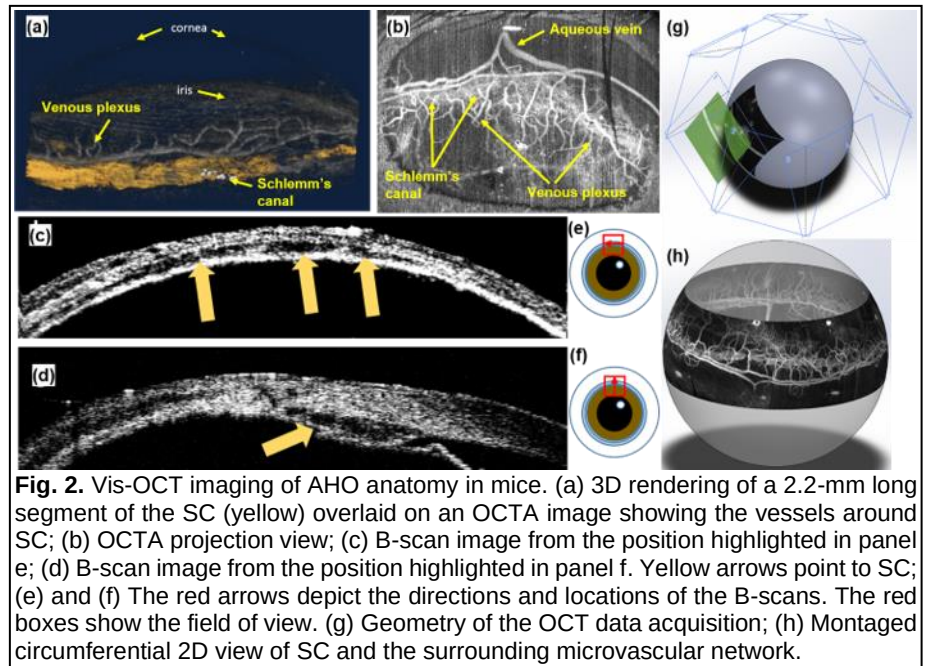


Fig. 2. Vis-OCT imaging of AHO anatomy in mice. (a) 3D rendering of a 2.2-mm long segment of the SC (yellow) overlaid on an OCTA image showing the vessels around SC; (b) OCTA projection view; (c) B-scan image from the position highlighted in panel e; (d) B-scan image from the position highlighted in panel f. Yellow arrows point to SC; (e) and (f) The red arrows depict the directions and locations of the B-scans. The red boxes show the field of view. (g) Geometry of the OCT data acquisition; (h) Montaged circumferential 2D view of SC and the surrounding microvascular network.

Balanced-detection supercontinuum OCT enables high-speed, shot noise-limited imaging in humans

Although supercontinuum lasers offer the needed flexibility in tuning OCT spectral range, supercontinuum laser-based OCTs suffer from an inherent high relative intensity noise (RIN). Excessive RIN introduces a spectrally-dependent background, especially at positions close to the zero-delay (Fig. 3a), added to other noises [37, 38]. If not removed, RIN reduces the SNR by up to 20 dB [39, 40], preventing vis-OCT from operating at the shot-noise limit and inducing a strong roll-off. The SNR constraint makes vis-OCT vulnerable to speckle and fundamentally limits the highest achievable A-line rate. For example, due to the limited allowable visible light exposure to the eye, vis-OCT only operated at an A-line rate of 25 kHz [41], with a strong RIN, which is approximately 8-fold slower than commercial OCTs using SLDs or swept source lasers. Such a limited A-line rate leads to motion artifacts and patient discomfort. We recently reported that balanced-detection vis-OCT using two well-calibrated spectrometers eliminated RIN's influence, improved the contrast-to-background ratio by 23 dB, and approached shot noise-limited operation [42]. As a result, the significantly suppressed RIN background enabled us to achieve a 125 kHz A-line rate in humans [42]. This balanced detection is similar to what is being widely used in swept-source (SS) OCT, except that SS-OCT performs balanced detection between two single-element detections instead of two calibrated spectrometers. SS-OCT often exhibits severe noise influence without balanced detection near the zero-delay position. We increased the A-line rate from 25 to 250 kHz for

high-speed imaging. Figs. 3a&3b compare two B-scan images of a tape roll, where the improvements in imaging contrast and depth are demonstrated by balanced detection. Figs. 3d & 3e show improved retinal imaging quality by balanced detection, setting the stage for the proposed high-speed AS-OCT. Based on the improved accuracy from balanced detection, we quantified the volumes of the anterior and posterior chambers [43], segmental SC size [44], and the locations of all CCs in rodents [45].

In summary, our preliminary data show that (1) human AHO is segmental, partially improving our understanding of its anatomy; (2) quantifying AHO's circumferential anatomy in mice is feasible; and (3) we established the technical foundation for high-speed, high-quality supercontinuum OCT in humans. These results set the stage for us to implement freeform robotic AS-OCT for clinical AHO anatomical imaging, to reconstruct the digital twin of AHO pathways, and to perform a novel AHO structure/function correlation evaluation in mice and humans by integrating AS-OCT with AA or AS-OCT angiography.

APPROACH

AIM 1: To develop, validate, and optimize a robotic AS-OCT platform

Rationale: An optimal way to image circumferential AHO anatomy must satisfy the following design constraints. An optimal spectral range is needed for the AS-OCT to balance sufficiently high axial resolution with adequate imaging depth without significant signal roll-off. Both axial and lateral resolutions and the B-scan to B-scan spacing must be $5\ \mu\text{m}$ or less to capture CCs [46]. Imaging must be done conformally to maintain normal incidence to AHO pathways for the highest imaging accuracy. For future clinical translation, imaging is best performed with the patient looking forward in a relaxed state for maximum repeatability. The AS-OCT must maintain a safe clearance distance from the eye and face for safety. The working distance must be long enough to avoid obstruction from the nose when imaging the nasal quadrant. Lastly, imaging must rapidly complete high-density scanning (~ 15 seconds per ocular quadrant) for patient tolerability and compliance. Building on our mouse AHO imaging experience (Fig. 2), we will design and optimize a new AS-OCT with emphases on (1) spectral range and imaging quality, (2) robotic control, and (3) patient safety.

Spectral range identification

According to the anatomical features of human AHO, the desired spatial resolution needs to be $<5\ \mu\text{m}$ to resolve CCs (as small as $10\ \mu\text{m}$) [46], and the penetration depth needs to be $\sim 500\ \mu\text{m}$ to reach the deepest aspects of SC in the limbal sclera. We conducted a Monte Carlo simulation using published tissue optical properties [47] to identify the optimal AS-OCT spectral range. If the bandwidth covers $1030\ \text{nm}$ to $1130\ \text{nm}$, we will have a minimum penetration depth of $1\ \text{mm}$ (a signal drop of $20\ \text{dB}$ for a double path of $500\ \mu\text{m}$) and an axial resolution of $5\ \mu\text{m}$ in air or $3.6\ \mu\text{m}$ in tissue. The depth of imaging using a 2048-pixel spectrometer will be $3.7\ \text{mm}$ in tissue. Adjusting the spectral

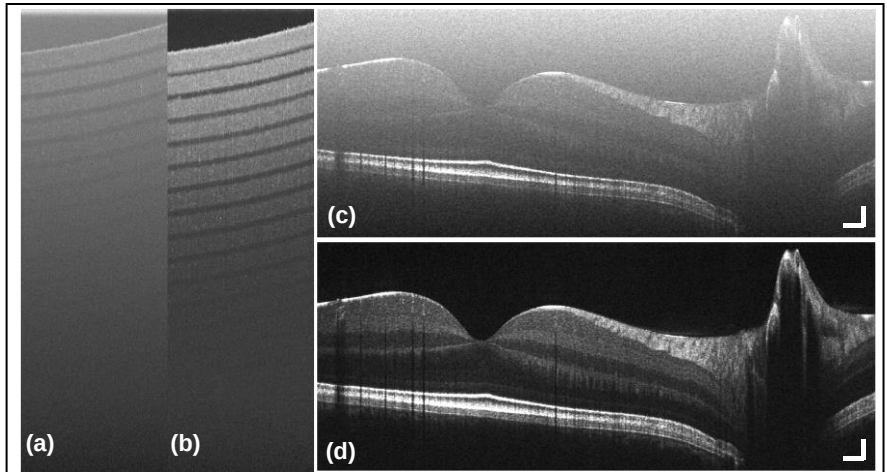
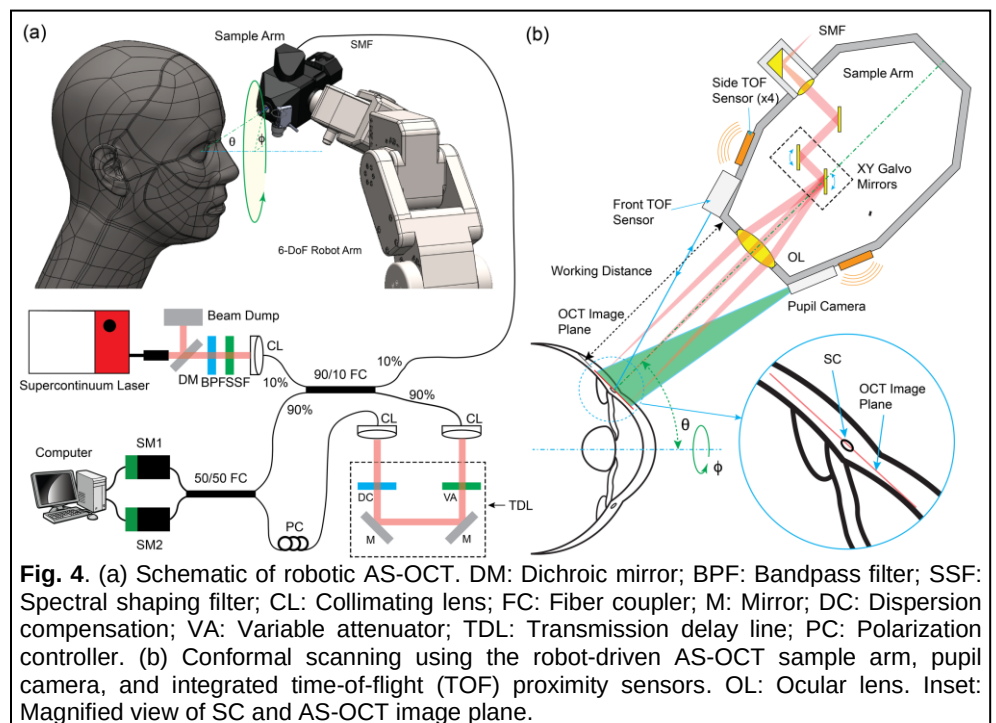


Fig. 3. Balanced-detection supercontinuum OCT. Tape roll imaged by (a) regular and (b) balanced detection vis-OCT; Human retina imaged by (c) regular and (d) balanced detection vis-OCT.



range is anticipated for optimization, and AS-OCTs for mice and humans require different spectral ranges, so we choose flexible supercontinuum lasers as the light source in the technology development stage. After optimization, we can potentially implement a swept-source AS-OCT.

Human robotic AS-OCT development

We have already developed a robotic AS-OCT for rodents [43-45] and have tested its ability to image the circumferential AHO pathways in wild-type mice and the AHO response to medications [44] (see Aim 2 for more details). The proposed human robotic AS-OCT will build upon our experiences with the rodent system to minimize potential risks. Fig. 4a shows the schematic of the robotic AS-OCT. The supercontinuum laser has an output covering 450 nm to 2300 nm. It will pass through a dichroic mirror, a bandpass filter, and a spectral shaping filter before coupling into the input arm of a 10/90 fiber coupler. The 10% output port will be connected to the sample arm mounted on a high-precision 6-DoF robotic arm (Meca500, Mecademic Inc, see quote) with 5 μ m repeatable position accuracy. The 90% output will pass through a transmission delay line before being coupled into a 50/50 fiber coupler. The interferogram will be detected simultaneously by two spectrometers (SM1 and SM2) at an A-line rate up to 250 kHz. The robotic arm will position the sample arm to perform circumferential imaging.

The conformal scan by the sample arm (Fig. 4b) introduces an oblique position normal to the limbus. This offers the shortest optical path length to image the AHO pathways and to measure SC cross-sections. Meanwhile, the sample arm's oblique position permits circular scanning along the rotation axis (highlighted in Fig. 4a) to achieve circumferential imaging. We will use a long focal length ocular lens ($f=100$ mm) to avoid a potential collision with the patient's face. At a galvanometer's angular accuracy of 30 μ rad optical angle, the corresponding spatial precision at the AS-OCT image plane is 3 μ m, sufficient for imaging CCs [46]. The 360° conformal imaging will be conducted step-and-scan (Fig. 2g) because galvanometer scanning is much faster than robotic arm positioning and offers higher phase stability than robotic motion.

The sample arm includes one pupil camera (OV9734, Omniview), one front-view laser time-of-flight (TOF) proximity sensor (LR-XH100, Keyence), and four side-view laser TOF proximity sensors (VL6180X, STMicro). The front-view TOF sensor and the pupil camera are co-registered with the center of AS-OCT's field-of-view (FOV). These sensors provide real-time feedback on the surroundings, establishing the AS-OCT calibration and focusing procedure and safety protocol. The human robotic AS-OCT will be developed at Northwestern and transferred to UCSD for human donor eye imaging.

AS-OCT calibration and robotic control imaging procedure

Performing 360° conformal imaging requires precise positioning of the sample arm around the eye (Fig. 5a). The AS-OCT hardware is pre-calibrated to ensure the co-registration of the following four reference points: (1) the aiming point of the front TOF sensor, (2) the center of the OCT FOV, (3) the center of the pupil camera FOV, and (4) the tool center position (TCP), which is the origin of the tool reference frame (TRF) being attached to the sample arm.

The calibration procedure will establish the absolute spatial coordination between the TRF and the eye reference frame (ERF) representing the spatial conformation of AHO pathways. Thus, we can scan the sample arm's TCP along the desired path (orange circular path in Fig. 5a) conformal to AHO pathways. The calibration consists of 3 progressive coarse-fine steps: (1) locating the pupil center by analyzing the real-time pupil camera image. This will provide a coarse alignment of the TCP to the eye's optical axis in ERF (Fig. 5b); (2) acquiring an AS-OCT B-scan along the optical axis. The B-scan image will be leveled to refine the calibration of the eye optical axis in ERF. We will estimate the eye size to generate the preliminary scanning path perpendicular to the optical axis (OA, Fig. 5c); (3) performing B-scans at 4 locations equally distributed along the preliminary scanning path (Fig. 5d). Analyzing the lateral and axial offsets

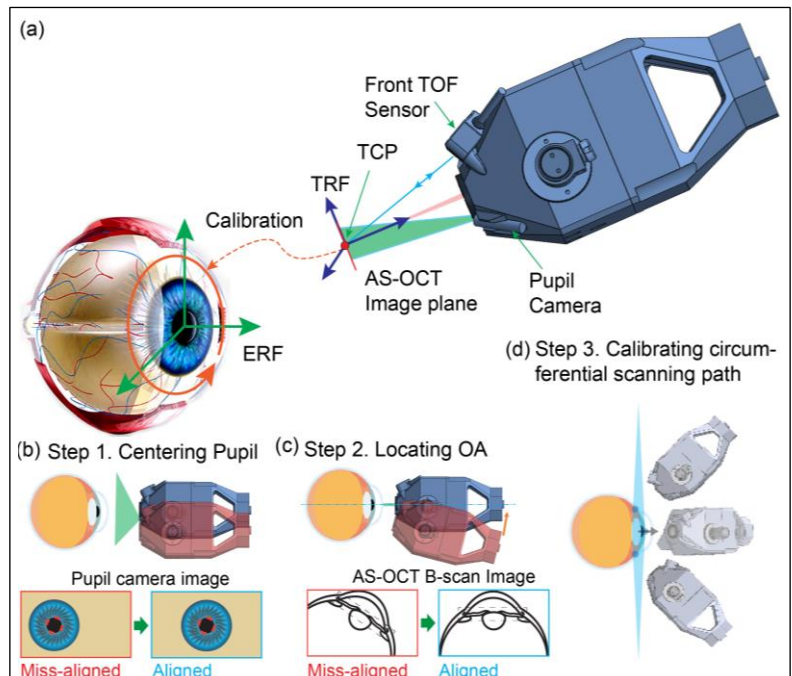


Fig. 5. AS-OCT robotic control. (a) Schematic illustration of the Tool reference frame (TRF) associated with the AS-OCT sample arm and the Eye reference frame (ERF). It allows the robot to scan the sample arm's Tool center position (TCP) along the desired orbit (orange path) conformal to the limbus, which includes these steps: (1) Using the pupil camera to locate the center of pupil (b); (2) aligning the optical axis of the sample arm to the ERF, using rapid AS-OCT B-scans (c); (3) rapid AS-OCT sampling at discrete locations to fine calibrate the spatial conformation of the eye (d).

of the B-scan images will finely adjust the shape of the scanning path, to accommodate the slightly elliptical shape of the human limbus. Finally, we will conduct a 360° scan along the calibrated path.

We will optimize and validate the robotic control to accommodate human facial features during AS-OCT imaging using phantom heads and numerical simulations. We will 3D print head phantoms specified by the National Institute for Occupational Safety and Health (NIOSH). NIOSH defined ten facial dimensions (age- and race-adjusted) with five face-size categories as the general representation for human facial anatomies [47] (Fig. 6a). The experimental configuration will also be digitally replicated in a commercial robotic simulator (RoboDK, RoboDK Inc.) featuring the dedicated driver for the Meca500 robot used in this study (Fig. 6a). We will use the RoboDK to program, simulate, and optimize the trajectory of the robot to avoid contact between the AS-OCT sample arm and the head phantom, which will define the safety clearance. The event will be recorded for further optimization if the clearance is within 1 cm (safety threshold). The optimized robot control in the simulation will then be validated by physical tests using 3D-printed face phantoms.

Human safety measures

We will develop a 4-tier redundant safety protection mechanism.

Tier 1: to enclose the AS-OCT sample arm with a soft polymer shell, such as Thermoplastic Polyurethane (TPU) and Elastic 50A photopolymer (Formlab Inc.). **Tier 2:** to monitor the real-time feedback from the pupil camera images. For a timely response, we will migrate the current CPU-based AI tracking to an AI-chip integrated camera (AI Sensor, MIIS Inc.). **Tier 3:** to establish a safety envelope using real-time feedback from the integrated TOF proximity sensors (Fig. 6b). When a potential contact is determined based on either the pupil camera or the TOF feedback, the control software will instantaneously stop the robot and turn off the laser. **Tier 4:** to enable a hardware safety button. During imaging, operators can press the interlock button to turn off the robot and laser for any concern.

Potential challenges and alternative approaches

We have already built a robotic AS-OCT for rodents (Fig. 8) [43-45]; hence, we do not foresee any engineering challenges in human system development. The Zhang lab has reported high-quality human retinal imaging using balanced detection vis-OCT [42]. Therefore, we do not anticipate any challenges in developing the proposed balanced-detection AS-OCT. In this project period, we will only focus on MIGS using human donor eyes; therefore, the proposed AS-OCT will satisfy all the technical needs.

AIM 2: To reconstruct digital twins of the anterior segments and establish the anatomy/function correlation of segmental AHO

Rationale

Creating the digital twin of each eye's AHO pathway anatomy will allow us to develop an eye-specific hydrodynamic model (see Aim 3 for details) to predict IOP reduction following MIGS, enabling personalized glaucoma surgical planning for the first time. Therefore, robotic AS-OCT image acquisition protocol and accurate 3D montaging are essential for establishing an accurate hydrodynamic model, as the model parameters depend on the anatomic details captured in the digital twin. Furthermore, examining the correlation between AHO anatomy in the digital twin and experimentally measured segmental AHO flow, combined with the hydrodynamic model, will provide more accurate surgical guidance. For example, we will be able to answer questions such as whether high- and low-flow regions correlate with the circumferential distribution of SC size and the number of CCs in each patient, providing guidance on MIGS [48].

Robotic AS-OCT image acquisition

In this project, we will only focus on data acquisition and 3D montaging for mice and human donor eyes. Future clinical imaging may require further optimization. Fig. 7 shows the data acquisition protocol. The AS-OCT image plane will be rotated around the limbus (Fig. 8a) to acquire eight

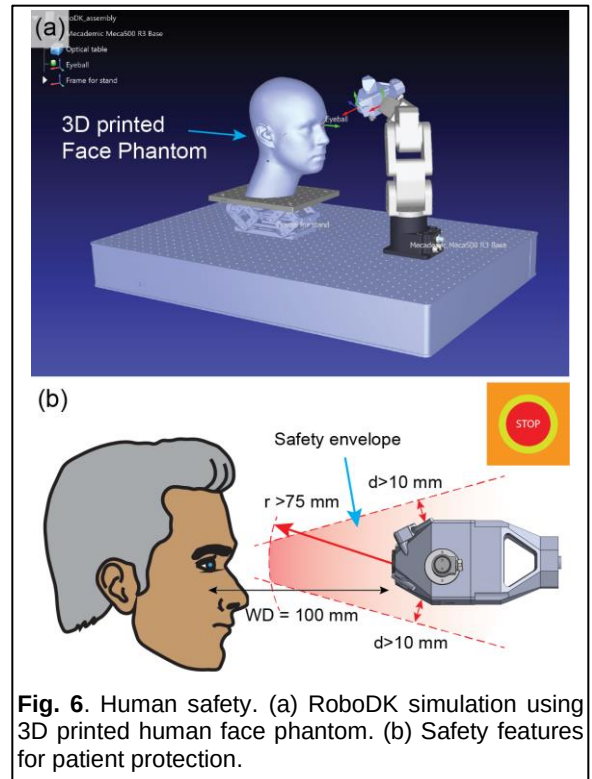


Fig. 6. Human safety. (a) RoboDK simulation using 3D printed human face phantom. (b) Safety features for patient protection.

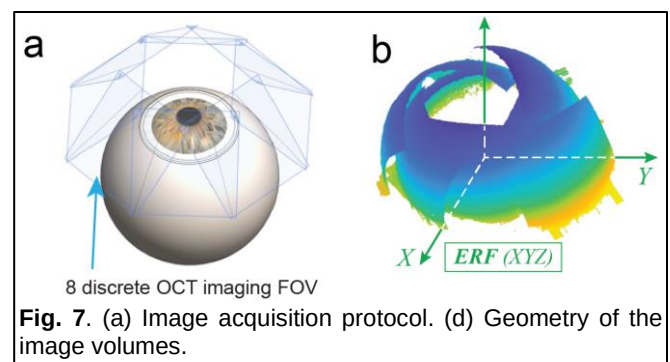


Fig. 7. (a) Image acquisition protocol. (d) Geometry of the image volumes.

independent volumes. Because the galvanometer offers much faster scanning speed and phase stability, positioning the robot at selected circumferential positions and acquiring multiple AS-OCT volumes yields higher imaging quality and stability than continuously rotating the robotic arm circumferentially [45]. The origin of the TRF is the focal point of the OCT (Fig. 5a), and the x- and y-axes represent the scanning directions of the galvanometer. In the ERF, the Z-axis is aligned with the visual axis, and the origin is located at the centroid of the trabecular AHO pathways. We will position the TRF around the Z-axis of the ERF at eight locations for AS-OCT acquisition with a 45-degree step size, where two adjacent volumes overlap 15% with each other. All eight volumes will be acquired with normal AS-OCT illumination. We will also document the TRF coordinates at each AS-OCT acquisition position for 3D montaging. Fig. 7b shows the directly assembled eight volumes based on the documented TRFs without registered 3D montaging in the ERF coordinate (Fig. 5a). It exhibits distinct artifacts of discontinuity and projection errors associated with data acquisition. To generate the digital twin reflecting the anatomical features, we will develop a registered 3D montaging method.

Registered 3D montaging

Figs. 8a and 5b show the workflow of 3D montaging inspired by the simultaneous localization and mapping (SLAM) method widely used in robotics and autonomous driving [49]. Our inputs are the robotic arm's TRF and the corresponding AS-OCT and AS-OCT angiogram (AS-OCTA) of the limbal vessel volumes. We will extract the top surface between two adjacent AS-OCT volumes and represent them as point clouds (step 1 in Fig. 8b). Between two AS-OCTA volumes, landmark point clouds will also be created (step 2). We will register the common landmarks between two adjacent AS-OCTA volumes. We will iteratively calculate the rotation and translation elastically (step 3) to complete the 3D montaging of the eight AS-OCT volumes (step 4) to reconstruct the 3D digital twin.

To demonstrate the feasibility of the proposed 3D montaging, Fig. 8c shows a cross-sectional image of the entire anterior segment extracted from the 3D digital twin, where the two arrows highlight the SC locations. The entire highly curved cornea and limbus are clearly imaged, and their total thickness is well beyond the imaging depth of the robotic AS-OCT itself. Then, we segmented SC circumferentially and visualized it with the corneal surface in Fig. 8d. We also found that, compared with the traditional AS-OCT imaging along one uniform illumination direction, conformal imaging improved the contrast-to-noise ratio (CNR) by more than five-fold, improved the repeatability of SC segmentation by nearly two-fold, and was sensitive to SC size variations induced by body posture [45].

Calibration and validating digital twin reconstruction

Because our hydrodynamic model's accuracy relies on accurately measured AHO anatomic parameters (see Aim 3), including SC height and CC size/location, we must carefully calibrate and validate our digital twin from 3D montaging. We will use the cornea surface as a reference for interactive 3D montaging of the AHO pathways. This enables us to apply the M-estimator Sample Consensus algorithm to determine the rigid geometric transformation between pairs of convex hulls [50]. We will calculate the coordinate transformation matrix T to align landmark points between adjacent AS-OCT volumes $\tilde{L}_n = [T]\tilde{L}_{n+1}$, where $\tilde{L}_n$ and $\tilde{L}_{n+1}$ are the coordinates of the convex hull points from AS-OCT volume n and volume $n+1$, respectively. Since rigid transformations are linear, the transformation can be applied sequentially to map all volumes onto the same ERF coordinate system (Fig. 7b). Once transformed, we will identify overlapping regions between adjacent point clouds and use the iterative closest point algorithm to refine the transformations relative to the ERF [51]. After correction, point clouds from all eight AS-OCT volumes will be aggregated into the common ERF, resulting in a smooth surface that highlights successful error correction compared with Fig. 7b.

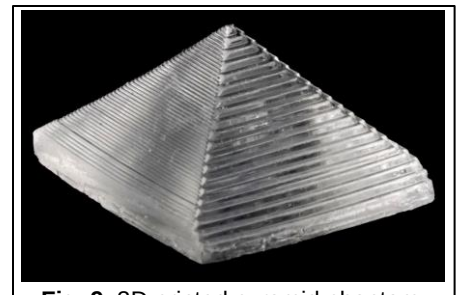
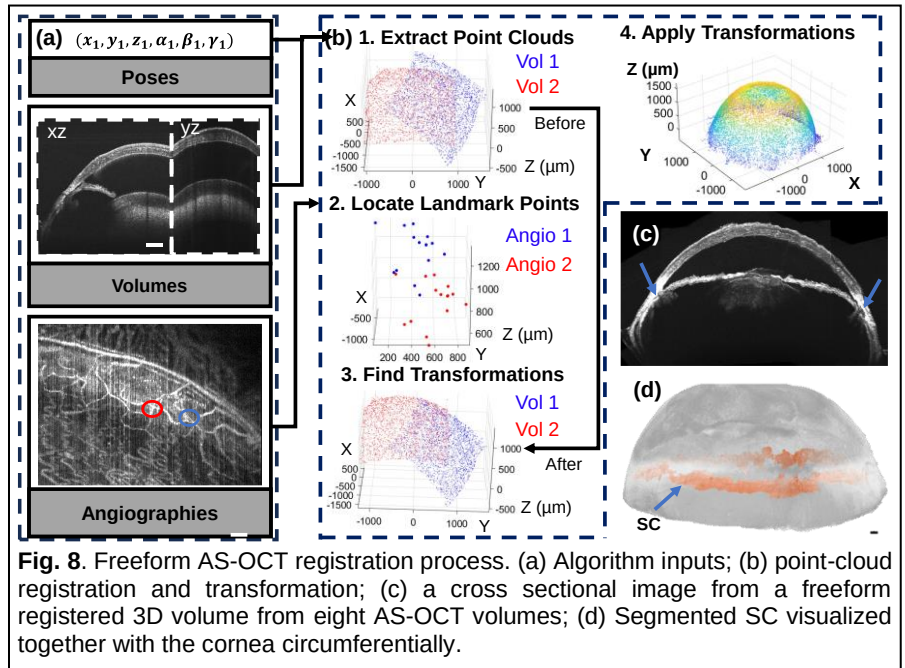


Fig. 9. 3D printed pyramid phantom.

We will validate the accuracy of the digital twin using both 3D-printed phantoms and transgenic mice. Fig. 9 shows the proposed pyramid phantom design, which offers asymmetrical features better than a spherical phantom for calibration purposes. The pyramid's four sides will have steps with different heights with a metal coating to maximize AS-OCT signal. We will vary the step size to examine the smallest resolvable feature size. Any 3D montaging errors will exhibit distorted or broken lines of the steps, which will be used to optimize our montaging algorithm. We will also calculate the volumes of the entire pyramid or volumes of different pyramid sections from the digital twin, which will be compared with volumes calculated from electron microscopy measured geometry [52].

We will image *Prox1-EGFP* transgenic mice [53] for *in vivo* digital twin validation. *Prox1-EGFP* mice have natively fluorescent SC [54-58], and can be imaged *in vivo* or *in ex vivo* flat mounts, as shown in Figs. 10a&10b. We will use our rodent robotic AS-OCT to obtain the digital twin of the AHO and circumferentially segment the SC, as shown in Fig. 8d. Meanwhile, we will build a scanning fluorescence microscope (with a lateral resolution of 4 μm and the same working distance as AS-OCT) to image the SC EGFP fluorescence from different well-controlled angles within the ERF. To compare the 2D fluorescence microscopy image with the 3D digital twin, we will create a project 2D image from the digital twin along the same angle as the fluorescence microscopy in the ERF. The 2D projection image shall reflect the same anatomical features as the fluorescence image.

We will validate the accuracy of the digital twin under drug-induced SC alterations to test its ability to detect AHO anatomical changes. Muscarinic agonists are known to lower IOP in mice and humans by contracting the longitudinal ciliary muscle, pulling down the scleral spur, and opening SC [59]. We tested the effect of applying 1% pilocarpine (also a muscarinic agonist) in wild-type mice. Using the circumferential SC segmented from the digital twin (Fig. 8d), we compare the regional SC volume in a polar plot (Fig. 10c) before and after 1% pilocarpine and show the percentile change in the polar plot in Fig. 10d. For the first time, we can visualize focal SC regional volume and regional volume % alterations in response to pilocarpine application *in vivo* [44], supporting our goal to achieve patient-specific MIGS because both AHO anatomy and its response to medication are segmental. This result suggests we can create digital twins before and after MIGS to further evaluate the surgical itself and its outcome. Hence, we will repeat the SC size comparison between the project 2D image from the digital twin and fluorescence microscopy in *Prox1-EGFP* mice before and after pilocarpine to validate the accuracy of the digital twin in detection minute alterations.

To estimate sample size, our rodent robotic AS-OCT has an axial resolution of 1.3 μm and a lateral resolution of 7.1 μm [45]. The typical width of mouse SC ranges from 50 μm to 140 μm [45]. If we apply the student T-test to compare the digital twin projection and fluorescence microscopy measured SC width with 5% uncertainty, we need to image 16 mice to reach a significant level of 0.05 and a power of 0.95. Including the pilocarpine studies, we will image 32 *Prox1-EGFP* mice in total.

Correlating digital twin with AHO functions in human donor eyes

We will reconstruct and evaluate AS-OCT digital twins for human eyes, which will be performed at UCSD. Post-mortem human eyes will be set up in an anterior segment perfusion system. Using a constant-flow method (BSS at 2.5 $\mu\text{l}/\text{min}$), stable IOP and outflow facility (C) will be established at baseline [29, 60]. Robotic AS-OCT imaging will then be performed to circumferentially map AHO pathways around the human eye limbus. The anatomical positioning and dimensions of SC and CCs will be visualized and used to create the 3D digital twin following the method described in Fig. 8. Variability in SC dimensions and CC location and dimensions will be compared to histology (the structural gold standard) at sampled locations after H&E stain, using the same eyes in the experiments below.

The digital twin will be compared to AA that was simultaneously acquired from the same eye. AA is the gold standard for visualizing segmental flow using a patient-compatible method. AA involves placing fluorescein or indocyanine green (ICG) in the anterior chamber followed by ocular surface fluorescence imaging (Spectralis HRA+OCT; Heidelberg Eng.) (Fig. 1). In post-mortem (pig [61], cow [62], dog [63], cat [64], and human [65]) and live (non-human primates [66] and humans with [28] and without glaucoma [26, 67]) eyes, AA has shown segmental post-limbal AHO with LF and HF regions. By comparing the segmental AHO pathway (Fig. 10c) LF

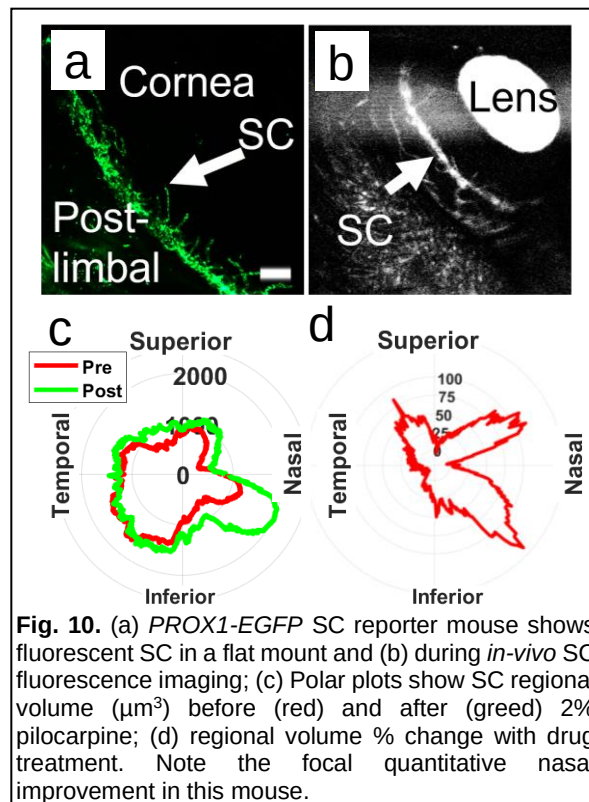


Fig. 10. (a) *PROX1-EGFP* SC reporter mouse shows fluorescent SC in a flat mount and (b) during *in-vivo* SC fluorescence imaging; (c) Polar plots show SC regional volume (μm^3) before (red) and after (green) 2% pilocarpine; (d) regional volume % change with drug treatment. Note the focal quantitative nasal improvement in this mouse.

and HF SC and CC parameters from the digital twin in the same eye, we can identify structural biomarkers that define segmental AHO that can then be used to identify LF and HF regions using non-invasive imaging only. Power analysis based upon variation in OCT-measured SC size in human eyes estimates needing six human eyes for this comparison [68].

Next, the digital twin will be compared to the AS-OCTA, which was simultaneously acquired. This unique approach bridges structural and fluid flow assessment in one setting. The same AS-OCT image can be used to create the digital twin (structural evaluation) and AS-OCTA image (fluid flow assessment). However, as we have published for both the anterior and posterior segments [69-72], OCTA needs a contrast agent, usually blood flow. Post-mortem ex-vivo eyes have no blood flow. Thus, we use FDA-approved and pharmaceutical-grade lipid emulsions (LE; 20% Intralipid), safe for intravenous delivery as an OCTA motion contrast agent [73, 74]. Thus, robotic AS-OCT and AS-OCTA will be performed around the limbus with the assistance of LE. More than the AA experiment, the presence of LE and use of AS-OCT will facilitate 3D montaging. AS-OCTA volumes will be analyzed using MIPAV software to determine the presence or absence of SC, CC, and distal pathways. Preliminarily, we show that both HF and LF regions can demonstrate variable presence of SC and CC. This is important because all LF and HF regions may not be equivalent, and understanding subtypes (based on SC, CCs, and the hydrodynamic model) may identify the best site for surgery. Similar to the above, we expect to need six human eyes for this comparison.

We expect to create anterior segment digital twins using robotic AS-OCT in both in vivo mouse eyes and ex-vivo human donor eyes. Variations in SC and CC parameters in the digital twin will match fluorescence imaging and histology. As a natural outcome of the digital twin, we can establish the baseline of the AHO anatomy, such as statistical variations of SC size and circumferential distributions of CCs. Such knowledge will help us guide MIGSs through building eye-specific hydrodynamic models.

Potential challenges and alternative approaches

We anticipate two challenges. First, because AS-OCT and fluorescence microscopy of the *Prox1-EGFP* mice will be performed sequentially, magnifications and imaging geometries between the two modalities may not be perfectly aligned for direct comparison. Although we believe that the digital twin itself will take care of the geometry mismatch because we can freely adjust the projection angle from the digital twin, potential deformation of the mouse eyeball can cause an intrinsic mismatch. We can leverage AS-OCTA and take advantage of the blood backflow into the SC to visualize the SC (Fig. 2b) simultaneously with AS-OCT acquisition [30]. Although blood backflow alters the SC morphology, such a change will not affect digital twin validation thanks to acquiring AS-OCT and AS-OCTA simultaneously. The second challenge is AS-OCTA in human donor eyes, where no blood contrast exists; however, we need AS-OCTA as the common landmark to create the digital twin (Fig. 8a). As described above, injecting Intralipid will create an artificial contrast for AS-OCTA to enable digital twin reconstruction. Finally, we note that variations in regional flow can be caused not only by variability in CC distribution and size, and SC area distribution, but also by variability in the hydraulic conductivity of the TM. While we cannot directly visualize this, our modeling described in Aim 3 will allow these differences to be characterized.

Aim 3: To develop an eye-specific hydrodynamic model to guide MIGS

Rationale

Sophisticated neurosurgeries, cardiovascular surgeries, and cancer treatment have long used 3D imaging of patients from radiological imaging (MRI and CT) for surgical planning and outcome prediction, but ocular surgical planning still relies on 2D *en-face* or cross-sectional images. In particular, ophthalmologists perform MIGSs without prior knowledge of a patient’s AHO pathways and have no tool to predict the outcomes. Obtaining the digital twin of a patient’s AHO pathways provides a golden opportunity to develop a hydrodynamic model to incorporate each patient’s unique AHO anatomy and analyze IOP reduction based on different numbers and locations of focal TM bypasses; hence, optimal MIGS outcome can be achieved. This aim will develop and optimize such a hydrodynamic model and validate it in mouse and human donor eyes.

Eye-specific hydrodynamic model development

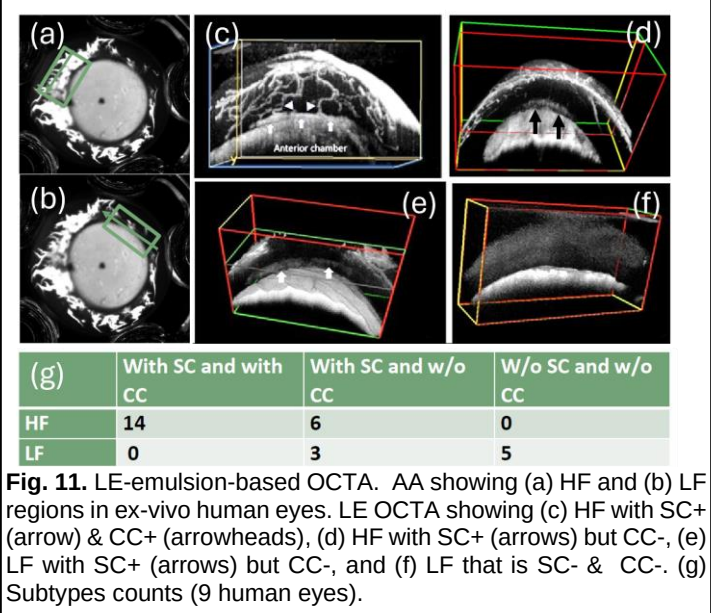


Fig. 11. LE-emulsion-based OCTA. AA showing (a) HF and (b) LF regions in ex-vivo human eyes. LE OCTA showing (c) HF with SC+ (arrow) & CC+ (arrowheads), (d) HF with SC+ (arrows) but CC-, (e) LF with SC+ (arrows) but CC-, and (f) LF that is SC- & CC-. (g) Subtypes counts (9 human eyes).

We recently developed a hydrodynamic model using a circuit network [25] (Fig. 12a). We used nodal voltages as proxies for aqueous humor pressures and current in the circuit branches as proxies for aqueous humor flow. We treated the flow resistance associated with TM, SC, and CC as electrical resistors in the circuit network. R_{CC} decreases as a function of IOP- P_{EV} (P_{EV} is the episcleral vein pressure), R_{TM} increases in glaucoma [9], and R_{SC} varies with local SC height. In the initial work, we used 30 CC resistors (R_{CC}), 30 TM resistors (R_{TM}), and 40 resistors in SC (R_{SC}) between each adjacent CC pair. These values reflected the literature report and represent a trade-off between accuracy and computational expenses [25]. Figs. 12b&12c respectively illustrate the AHO in the model before and after surgical intervention, where a theoretical stent was implanted to increase the outflow facility. In the model, we set the R_{TM} and R_{SC} values to zero at the corresponding stent locations and used Kirchhoff's rules to calculate the current in each branch. Model validation was done by comparing model prediction to literature-reported IOP values before and after stent implantation for various stents and situations; the agreement was good. [25]. Therefore, our model has the potential to predict MIGS outcomes if tailored for specific eyes.

In the current model [25], we assumed R_{SC} , R_{TM} , and CCs distribute uniformly circumferentially. Although this model worked well in population-based studies, these assumptions do not hold in individual eyes. Fortunately, we can extract eye-specific parameters from our digital twin to tailor the model. We will fine-tune R_{SC} values based on the local SC height values from the segmented SC morphology. Reduced SC height leads to an increased R_{SC} . We will also redistribute the position and size of CCs based on the digital twin. These two customizations will reflect the nature of segmental AHO in each eye. We will continue to distribute R_{TM} uniformly in our model because we cannot measure TM variations noninvasively for each eye.

We tested the feasibility of the model predicting IOP reduction under specific CC distributions. We obtained the polar distribution of CC locations in a human donor eye [75] and adjusted the model accordingly. We numerically placed one stent at varying polar coordinates and compared the IOP reduction against even CC distribution. As shown in Fig 13, IOP reduction varies from 4.2 mmHg to 10 mmHg, depending on the stent location. The next step is to validate the model prediction in mouse and human donor eyes after surgical manipulation. We will implement the hydrodynamic model as a web application, and the source code will be deposited at the Zhang lab's GitHub page <https://github.com/FOIL-NU> for public access.

To test the hydrodynamic model in mice using AHO pathway laser ablation

Because performing trabecular bypass surgery in mice is not feasible, we will conduct AHO pathway laser ablation as previously reported [76] to introduce AHO perturbation. Using a 532 nm continuous-wave diode laser (50- μ m spot size, 80-mW illumination power, and 150-ms illumination duration), we will perform laser ablation at selected circumferential locations around the grey zone of the limbus. We will use wild-type C57BL6/J mice in this study. We will use an iPerfusion system [77, 78] to measure the outflow facility of the mouse eyes and use the published rate of production of aqueous humor in mice [79] to adjust R_{TM} , R_{SC} , R_{CC} , and P_{EV} so that our hydrodynamic model fits mouse eyes. Then, we will obtain the digital twins of the mouse eyes using the robotic AS-OCT and document the IOPs before and after laser ablation targeting TM, SC, and CC. The CC distributions from the digital twins will be used to tailor the model. We will also redistribute the R_{SC} values to be correlated with the local SC height as described in [25]. The digital twins before and after laser ablation will inform the model of the location of the TM ablation, where we will significantly increase the local R_{TM} value and eliminate any CCs of those elements in the model. We will also seek to adjust R_{CC} based on the image CC lumen dimension rather than using a uniform value.

We estimate the sample size based on the simulated model sensitivity, where the expected range of IOP alteration is 16.8% to 40% (Fig. 11). A commonly used Tonopen has an accuracy of ± 2 mmHg, and a baseline IOP of wild-type C57BL6/J mouse is 11 mmHg. If we want to detect a 16.8% IOP reduction in the mouse eye (9.2 mmHg), we need 34 mice to compare before and after laser ablation, assuming 95% power and a two-sided significance level of 0.05. Assuming a 90% successful laser ablation rate, we will image 40 mice.

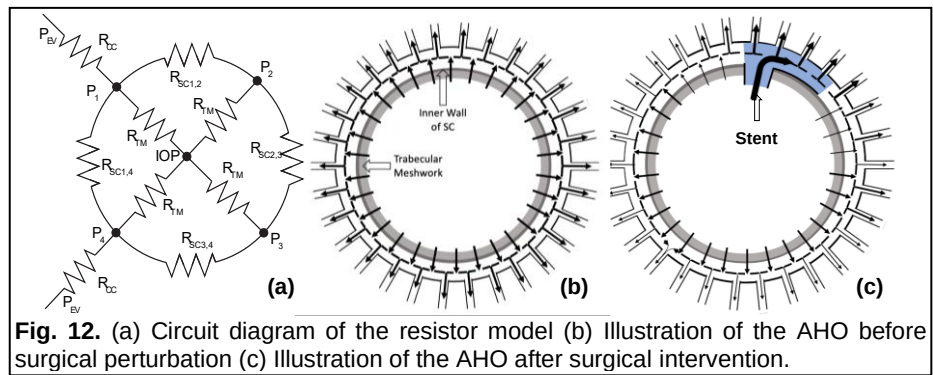


Fig. 12. (a) Circuit diagram of the resistor model (b) Illustration of the AHO before surgical perturbation (c) Illustration of the AHO after surgical intervention.

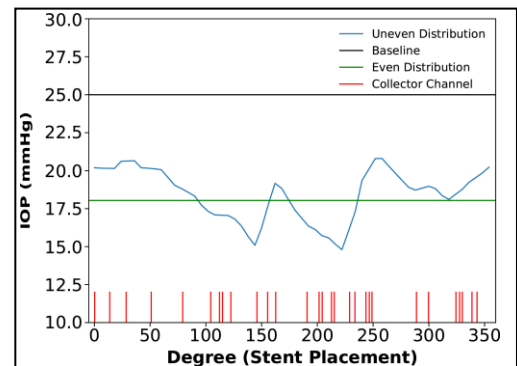


Fig. 13. Model prediction of IOP reduction with CC distribution given by the digital twin.

To test the hydrodynamic model in human donor eyes with MIGSs

We will also evaluate our hydrodynamic model in human donor eyes to facilitate translating these results to glaucoma patients. The simple fact of segmental AHO means that not all AHO pathway regions are equivalent [26, 67]. Therefore, surgery in specific AHO regions could lead to better results. Already, AA in glaucoma patients shows that trabecular MIGS in baseline low-flow (LF) regions demonstrate focal AHO rescue [28]. In the lab, AA has also been used to visualize LF and HF areas in *ex-vivo* pig and human

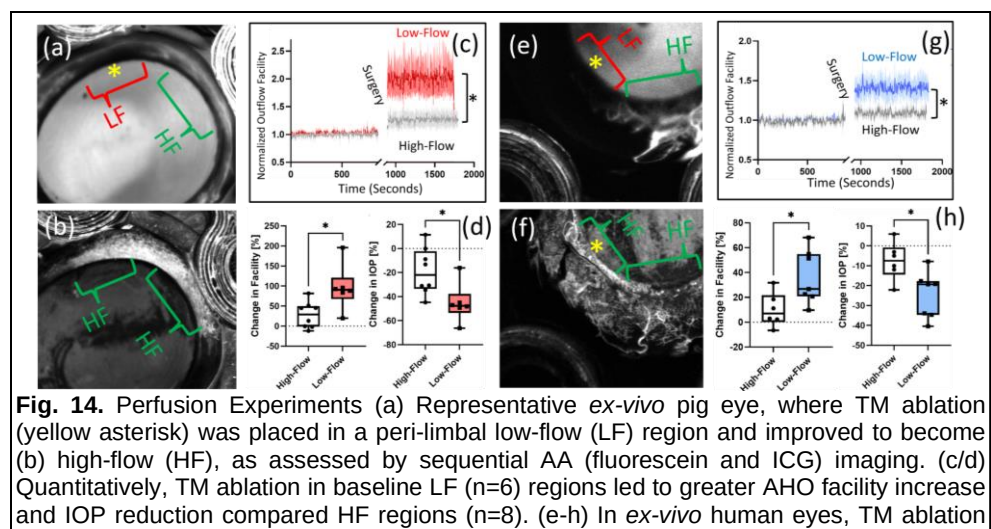


Fig. 14. Perfusion Experiments (a) Representative *ex-vivo* pig eye, where TM ablation (yellow asterisk) was placed in a peri-limbal low-flow (LF) region and improved to become (b) high-flow (HF), as assessed by sequential AA (fluorescein and ICG) imaging. (c/d) Quantitatively, TM ablation in baseline LF (n=6) regions led to greater AHO facility increase and IOP reduction compared HF regions (n=8). (e-h) In *ex-vivo* human eyes, TM ablation

After setting up stable baseline IOP (mmHg) and outflow facility “C” ($\mu\text{L}/\text{mmHg}/\text{min}$), TM ablation MIGS was performed in LF and HF regions with IOP and C measured again. A greater increase in C and reduction in IOP was seen in surgery targeted to baseline LF regions (Figs. 14d & 14h). While these results support a strategy of targeting TM MIGS, the problem is that AA is laborious, time-consuming, and invasive. Therefore, TM bypass MIGS targeting will be evaluated using the digital twin and hydrodynamic model in the human anterior segment perfusion system to develop *non-invasive* surgical guidance.

Human donor eyes will be obtained from San Diego Eye Bank within 24 hours of death. Eyes with ocular disease other than having cataract surgery will be rejected. Eyes will be set up in an anterior segment perfusion system (iOnly Human) [29]. Using a constant-flow method (BSS at 2.5 $\mu\text{L}/\text{min}$), stable IOP and C will be established at baseline. AS-OCT will be performed to circumferentially map AHO pathways around the limbus to create the digital twin. The anatomical positioning and dimensions of SC and CCs will be used to create a customized hydrodynamic model for each eye. TM bypass surgery will then be targeted to locations predicted to have the least, partial, and greatest impact on C based on the hydrodynamic model. A trabecular microbypass stent (Glaukos) will be used, as previously published [65]. This will be done by unclamping and inverting the eye under a surgical microscope (Topcon OMS-90). After surgery, the eye will be re-clamped with perfusion restarted. Leaks will be identified by testing for fluid flow around the base of the clamped sclera. Power analysis based on our prior work (Fig. 14) [29] estimates that we need ten eyes per group. IOP and C will be re-measured and compared to before surgery. After surgery, robotic AS-OCT, AA, and histology will be performed to validate that the TM bypass MIGS was properly performed.

After TM bypass MIGS, we expect to see full-thickness TM opening from AS-OCT and histology. AA will show fluorescent tracer flow in the post-limbal area adjacent to the TM bypass location. We hypothesize that the greatest improvement in C and reduction in IOP will occur in surgery targeted to AA LF regions, which will be correlated with the hydrodynamic model.

Potential challenges and alternative approaches

As mentioned above, we cannot directly visualize the variability in the hydraulic conductivity of the TM noninvasively. However, using our model post-surgically in human donor eyes and post-laser in mice eyes, we will be able to evaluate the role that TM variability plays in regional flow, which characterizes the significance of this variable. While our idea is that targeting baseline LF regions will result in greater improvement in C and reduction in IOP, variability was seen in our original results [29], and this conclusion may be too simplistic. Thus, our current approach is effectively an unbiased alternative, as the digital twin and hydrodynamic model direct the location of TM bypass. Even if our original hypothesis is incorrect, if the current targeting strategy yields stronger and more consistent success, we still have a method for personalized therapy in glaucoma patients.

Biological variables

While it is known that IOP increases with age due to the lack of well-controlled studies, the role of gender is more complicated. For instance, while some work has shown increased age-controlled IOP in women compared to men [80], this effect becomes non-significant when the time of day (a known variable) is strictly controlled for [81]. Regarding race, Aim 1 safety testing uses NIOSH data that includes input from multiple racial and ethnic strata [47]. Ultimately, we will not exclude any subject or eye based on sex or race for any experiment. We will report the sex, age, and race of all human donor eyes and the gender and age of all mice imaged.

Rigor, reproducibility, and transparency

The proposed research builds upon our long track record of developing supercontinuum OCT for ophthalmic applications. The Zhang lab reported the first supercontinuum OCT retinal imaging in 2013 [31, 82] and has continuously pushed its high-resolution and functional imaging applications in both rodents and humans [32, 39, 41, 82-92]. The Zhang lab also reported the first *in-vivo* supercontinuum OCT imaging of AHO pathways in rodents in 2020 (Fig. 2) [30], which builds the foundation of this proposed work. The successful demonstration of the rodent robotic AS-OCT (Fig. 8) minimizes the risk of developing the human robotic AS-OCT. In addition, structural and functional AHO imaging using AS-OCT, AS-OCTA, and AA will be further validated by quantitative outflow facility measurements for the first time. All rodent experiments will use the fellow eye as the intersample control for anatomy/function correlation studies. Most rodent experiments will be cross-validated by results independently obtained by the UCSD and Northwestern teams. For *in vivo* rodent studies, we will emphasize rigor and reproducibility, including AHO pathways measurement test-retest repeatability assessment.

Most importantly, the rigor of our prior work is best represented by the multiple groups developing supercontinuum OCTs, resulting in high-quality papers published by researchers worldwide. After we reported anatomical and functional imaging using supercontinuum OCT, over ten other groups independently developed and reported supercontinuum OCT anatomical and functional imaging in rodents and humans [93-108]. This proves that supercontinuum OCT technology is reproducible and cross-validated. Recently, NIH Common Fund Oculomics program selected one project to investigate retinal biomarkers associated with Althermize’s disease using supercontinuum OCT or visible-light OCT, demonstrating the clinical potential of this technique.

Statistically, all sample sizes have been calculated using power analyses based on preliminary results or published data, which provide insight regarding the experimental question (see Vertebrate Animal). All experiments will also include proper controls. Experiments comparing HF and LF flow regions act as controls for the other. All reagents will be authenticated as separately described. Analyses will use two-way ANOVA, with Benjamini-Hochberg correction for false discovery, to assess significance. Statistical guidance will be additionally acquired from the Biostatistics Module staff at the Shiley Eye Institute NIH P30 core. Researchers in this project will be required to present their processed data, along with raw data sets, at regular group meetings. Critical data sets will be verified openly and independently by additional researchers. Moreover, our data and progress will be reviewed rigorously by meetings between Drs. Zhang’s and Huang’s teams on a regular basis via teleconference, email, or in-person meetings.

Project timeline and impact on science and clinical care

Table 1 shows the project timeline. All aims will start in parallel at both Northwestern (NU) and UCSD. After the initial test, the human robotic AS-OCT will be installed at UCSD at the end of Y2. Safety protocol development and validation will be performed at UCSD under clinical expertise starting with 3D-printed human heads. Transgenic mice will be transferred from UCSD to NU to establish a colony under an MTA agreement. Data acquisition and processing protocols to reconstruct the digital twin will be completed at NU. Animal validation and human donor eye validation of the digital twin will be conducted in parallel at NU and UCSD. Investigators from both NU and UCSD will collectively develop and optimize the hydrodynamic model throughout the project period. Animal and human donor eye validation of the hydrodynamic model will be performed in parallel at both NU and UCSD.

Tasks (Site)		Y1	Y2	Y3	Y4	Y5
Aim 1	Human robotic AS-OCT development (NU)					
	Robotic AS-OCT safety protocol development (NU)					
	Robotic AS-OCT safety protocol validation (UCSD)					
Aim 2	Robotic AS-OCT image acquisition protocol (NU)					
	Phantom validation (NU)					
	3D montaging devel opment (NU)					
	Animal validation of the d igital twin (NU)					
Aim 3	Human donor eye val idation of the digital twin (UCSD)					
	Hydrodynamic model development (NU & UCSD)					
	Animal laser surgery validation of the model (NU)					
	Human donor eye MIGS validation of the model (UCSD)					

Table 1. Project timeline and performing site.

The proposed work will lead to several potentially transformative impacts on the fundamental investigation of AHO pathway in rodent models and clinical glaucoma care. First, the robotic AS-OCT will enable researchers to visualize the entire AHO pathways for the first time. The AHO pathway’s *in vivo* circumferential anatomic features, including segmental SC size distribution, the total number of CCs and their circumferential distribution, have not been accessible to researchers before. Second, the concept of a digital twin of the eye is principally new, providing new ways to visualize and track ocular anatomical alterations and allowing ophthalmologists to perform glaucoma surgical planning for the first time. Finally, integrating the hydrodynamic model with the digital twin allows for predicting the effects of different MIGS plans to achieve optimal patient outcomes, leading to personalized and precision glaucoma care. In addition, the hydrodynamic model deposited at our GitHub website also ensures a broad dissemination to the research community.